

Analysis of Automated Ball-Strike System and Strategy in Major League Baseball

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1 Abstract

In 2026, Major League Baseball introduced the Automated Ball-Strike system (ABS), which allows players to challenge calls on balls and strikes. Since the technology is new, the strategy is untested. This study investigates whether ABS challenges are based solely on pitch position or whether contextual factors also matter. Using pitch-level data from Baseball Savant, I analyzed both spatial features (distance from strike zone) and contextual variables to model challenge probability. The results show that spatial patterns remain consistent across the season, while contextual factors significantly affect whether a pitch is challenged. These findings suggest that challenge decisions are not purely location-based and that teams incorporate situational information when deciding whether to use a challenge.

2 Introduction

ABS allows players to challenge a ball or called strike. Pitchers, catchers, and batters are allowed to challenge the call, but they can not get any input from the dugout. Additionally, the challenge must happen immediately after the call. Players signal a challenge by tapping their head. To achieve a precise strike zone, players were measured between 10 AM and noon local time, with strict rules on clothes and posture. The strike zone is exactly 17 inches wide. The top of the zone is exactly 53.5% of the batter's height, and the bottom is 27%. An unintended consequence of this process was official heights of players drastically changed overnight. Teams only have two incorrect challenges; if they challenge a pitch and it is overturned they retain their challenge. However, if the call is confirmed, the team loses its challenge (Major League Baseball 2026).

Since the beginning of the Moneyball era of baseball, data has been analyzed. Since the ABS system is new, it brings new data to uncover trends and strategies. It is unclear whether challenges are driven by pitch location, game context, or both.

3 Data & Methodology

This data comes from Baseball Savant (MLB Advanced Media 2026), the home of all Statcast data collected by MLB. Savant allows users to make figures on the platform or download the raw data for their own analysis. This data was analyzed and plotted using MATLAB.

This data was downloaded from Savant. This data is from the beginning of the season until May 31. This includes about 3,600 challenged pitches and 25,000 eligible but challenged pitches from May 25 to June 6. Savant collects 118 statistics for each pitch, but the important ones are the x and z coordinates of the pitch, the top and bottom strike zone coordinates, and the contextual statistics like the count and inning.

Another important statistic is whether the pitch is a reasonable challenge. Baseball Savant defines a reasonable challenge as a pitch in which one of the following is true: the original call was incorrect, the pitch is within three inches of the strike zone edge and an overturn would gain at least 0.3 runs, or the pitch has an expected challenge rate of at least 20%. Baseball Savant helpfully calculates all of these statistics, except the distance from the zone, which was necessary anyway. Savant uses a model by Tangotiger to calculate run expectancy (Tango 2018) and expected challenge rate (Tango 2023). I filtered all the unchallenged pitches using these conditions to determine if an unchallenged pitch was reasonable. To calculate the distance from the strike zone edge, I adjusted the edge of the strike zone to account for the radius of ball. The given x,z coordinates of pitch are for the center of the ball, so adding 1.45 inches to each strike zone edge accounts for this. For just wide, high, or low pitches, I just found the distance between the edge and the pitch. For pitches in the corners, I used the Pythagorean Theorem.

The first section of data is the location data, split into two categories: position and distance from the strike zone. For the position analysis, I plotted the location of reasonable, confirmed, and overturned challenges and included the average strike zone for reference. This allows for comparison of challenge outcomes by position. It also gives a visualization of the distance the ball is from the strike zone. The distance data was used to plot the distance from the strike zone by date. This was used to determine trends both by date and outcome. I predicted that the average distance would decrease as the season progressed, as players got more experience with the ABS system.

The second section of the data is the contextual data: count, inning, outs when up, runners on when up, and pitches in the at-bat. Using data from the challenged sample, I plotted these variables against distance to analyze whether the context affects the accuracy of the challenge. I also plotted the probability of a pitch being challenged based on the context. This demonstrates the player's willingness to challenge a pitch based on the situation.

Finally, using the pitch type given by Baseball Savant, I graphed the average distance vs the pitch type. This shows whether specific pitch types cause less accurate challenges.

4 Results

4.1 Location Results

To determine whether challenge outcomes vary by pitch location, the locations of confirmed and overturned challenges were plotted relative to the strike zone.

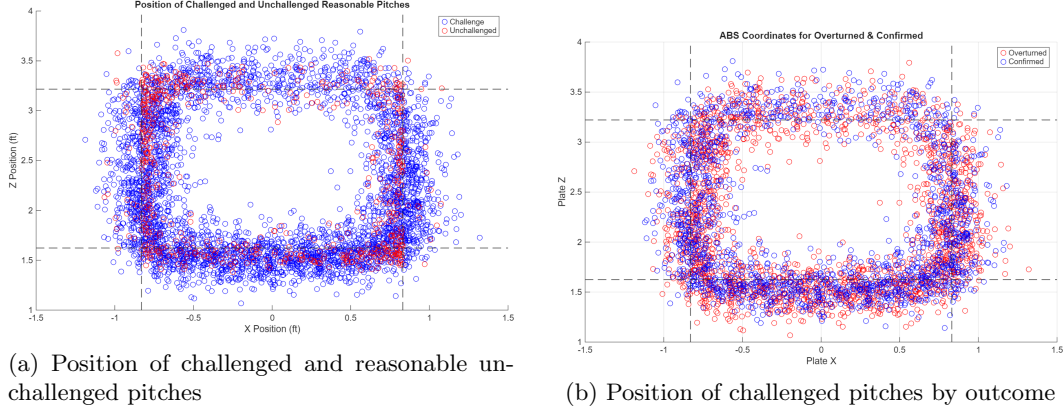


Figure 1: Comparison of reasonable unchallenged, confirmed, and overturned ABS challenges by location.

The reasonable unchallenged pitches are concentrated closer to the strike zone than the challenged pitches. Both types of pitch are denser on the bottom half of the zone. Both confirmed and overturned challenges are concentrated near the strike zone boundary. The two overlap, with no obvious region where one outcome is more dominant. These figures suggest that pitch location alone may not fully explain challenge outcomes.

4.2 Distance Results

To determine if players became more accurate as the season progressed, the distance from the strike zone edge was plotted by date.

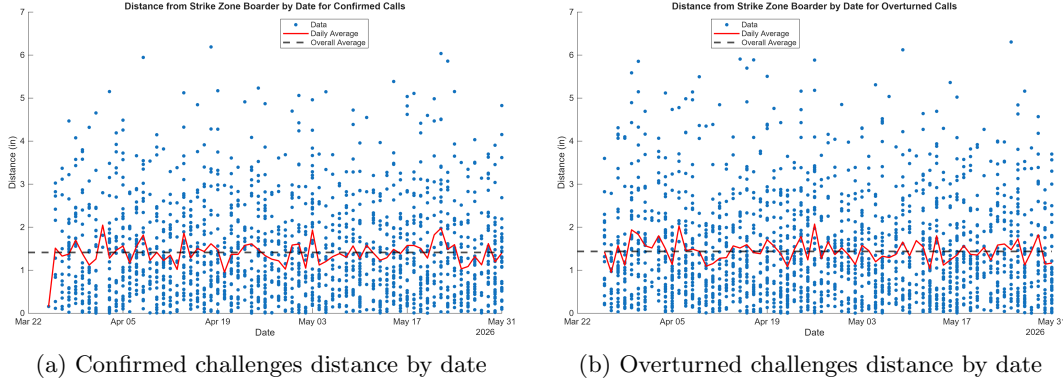


Figure 2: Distance by date by challenge outcome

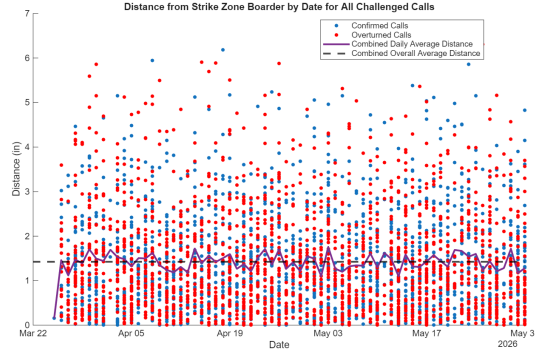


Figure 3: Distance by date by challenge outcome, confirmed and overturned

The average challenge distance remained relatively stable throughout the season. This holds for confirmed and overturned challenges. The distance distribution is more dense under the average. This suggests that distance is less influential than context.

4.3 Context Results

To assess the influence of game context on challenge behavior, both challenge frequency and average distance from the strike zone boundary were analyzed across innings, counts, base states, outs, and at-bat lengths. First, challenges relative to the count were examined to determine if a pitch's importance within the at-bat determines challenge behavior.

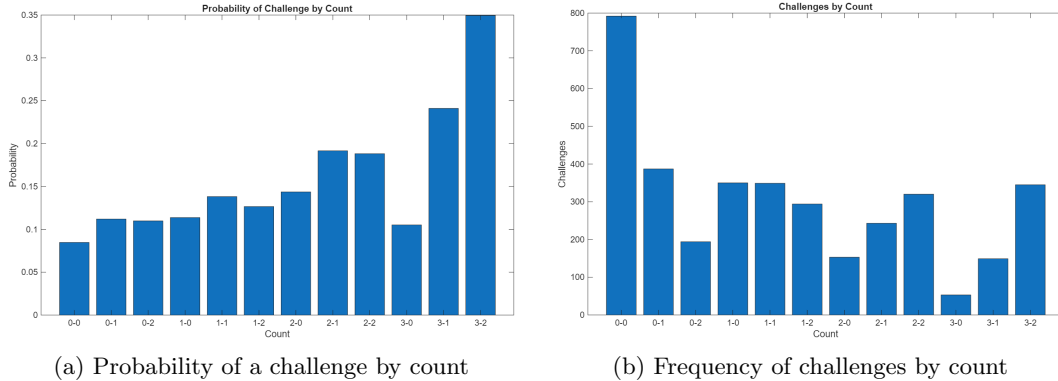


Figure 4: Challenges by count

The most frequent challenge occurs in a 0-0 count. However, the most likely to be challenged is a 3-2 count. This suggests that teams are most likely to use a challenge when the outcome of a single pitch immediately determines the outcome of an at-bat. To further investigate the role of context, challenge behavior was next examined as a function of the inning.

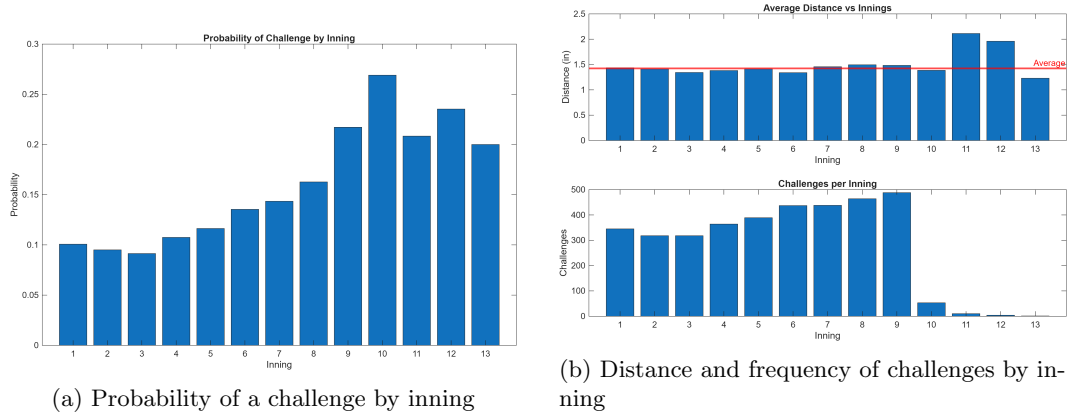


Figure 5: Challenges by inning

Pitches are most likely to be challenged in the later innings. The eighth and ninth innings are also the most frequent challenges. The average distance by inning is consistent, except after the tenth. This could be due to the small sample size, as MLB games are less likely to reach past the tenth inning. This indicates that players are more likely to challenge in high-leverage, late-game situations. Another important piece of context is outs when up. Thus, challenge behavior by outs was analyzed.

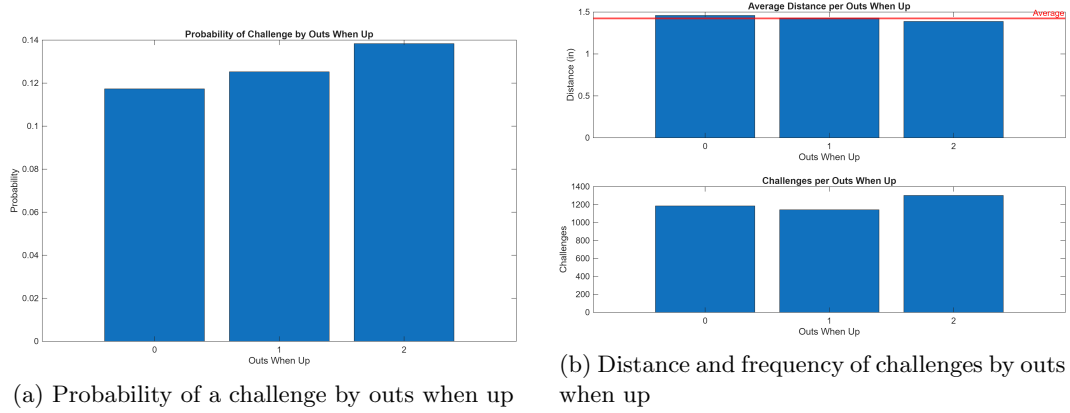


Figure 6: Challenges by outs

Challenges are most likely with two outs. The average distance was constant, suggesting little difference in accuracy. There is also not a significant difference between challenge frequency between outs. This suggests that outs may not be a significant factor in players' decisions to challenge a pitch. Likewise, runners on when up could influence a challenge decision.

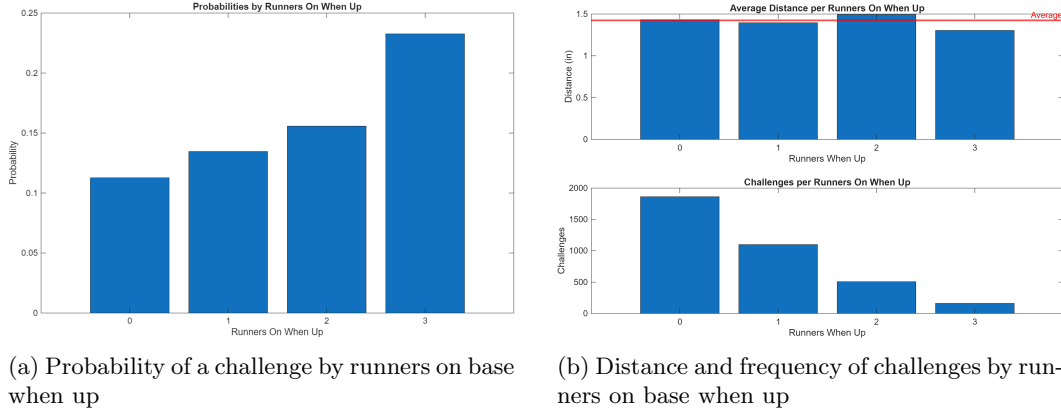


Figure 7: Challenges by runners

Challenges are most probable when the bases are loaded, twice as likely as if the bases are empty. Accuracy seems to be consistent with runners on base, except for when the bases are load. This could be explained by the small sample size. These findings indicate that runners on base during the at-bat are influential in challenge behavior. Finally, the number of pitches in the at-bat could change a player's willingness to challenge.

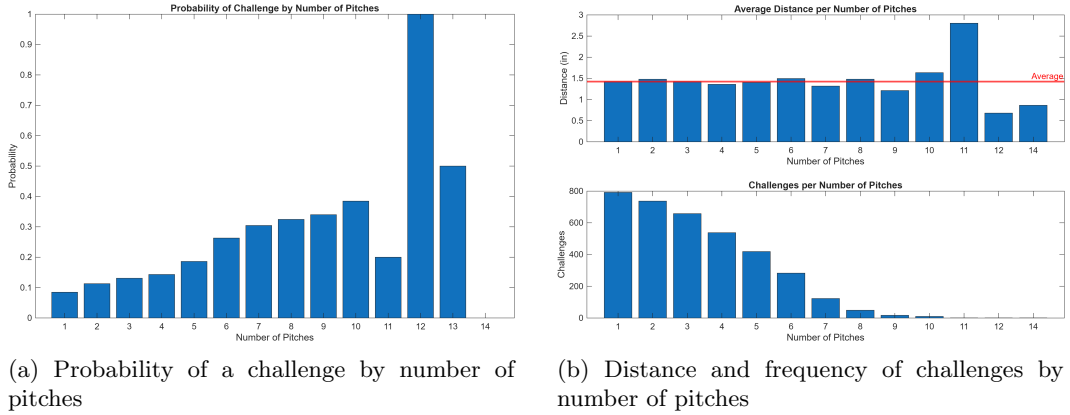


Figure 8: Challenges by number of pitches

The probability of a challenge increases with the number of pitches in the at-bat. Surprisingly, there is a 100% probability of the twelfth pitch being challenged. This can be explain by the low frequency of at-bats going twelve pitches, the low frequency of those pitches being taken, and the sample size of eligible pitches. The eleventh pitch has a much higher distance from the strike zone than average. This can also be attributed to a small sample size. This is consistent with context influencing challenges.

5 Discussion

Overall, the analyses indicate that ABS challenge decisions are influenced by both pitch accuracy and game context. Pitch accuracy is clearly an important factor, as challenges were concentrated near the strike zone boundary rather than on pitches located well within or well outside the strike zone. However, pitch accuracy alone does not fully explain challenge behavior. Analyses of several contextual factors revealed that challenge probability varied across different game situations, suggesting that players also consider the strategic importance of a pitch when deciding whether to initiate a challenge.

The importance of pitch accuracy is evident in the spatial distribution of challenged pitches. Challenged pitches are concentrated around the edge of the strike zone. A greater density of challenged pitches is also observed along the lower edge of the strike zone. This pattern may reflect the difficulty of judging low pitches or a greater willingness of players to challenge calls in that region (Figure 1a). In addition, few challenges are made far outside the zone and well within the zone (Figure 1b). This indicates that both players and umpires are generally good at calling pitches. Umpires are not making bad calls. Players are not challenging obvious balls and strikes. The average distance from the strike zone for confirmed (1.41 inches) and overturned (1.44 inches) challenges is close (Figure 2). The data indicate that the average distance is nearly identical between the challenge outcomes.

While pitch accuracy provides the foundation for challenge decisions, the context analyses indicate that it is not the only factor influencing when a challenge is used. If pitch accuracy were the only important factor, then the probability of a challenge would be constant across game situations. However, the observed data does not support this expectation. Challenge probability varies across count, inning, number of pitches in the at-bat, and base state (Figures 4a, 5a, 8a, 7a), suggesting that game context plays a meaningful role in challenge decision-making.

The hypothesis predicted that players would become more accurate in their ABS challenges over the course of the season. However, the results show that the average challenge distance by day remained largely stable, with no clear evidence of systematic improvement over time. This suggests that player accuracy did not meaningfully change throughout the observed period. One possible explanation is that players had already gained sufficient familiarity with the ABS system during spring training in both 2025 and 2026, which may have reduced any observable in-season learning effect. Additionally, the time window of the dataset may not have been long enough for a gradual learning curve to emerge. It is also possible that any improvement in accuracy is relatively small in magnitude and therefore not detectable at the aggregate level of this analysis.

There are several limitations in this analysis that may affect the interpretation of the results. The data is from March 25 to May 31 during the 2026 season, which represents only the early portion of the season. As a result, longer-term trends or strategic adjustments may not yet be observable. Additionally, the unchallenged pitches are only from May 25 to June 6 due to the size of the data. Although this subset contains 25,000 pitches and is therefore large in absolute terms, it may still not be fully representative of the broader distribution of unchallenged pitches across the entire season. This mismatch in time windows could introduce bias when comparing challenged and unchallenged pitches. While this difference may affect the absolute probability estimates, the relative comparisons between groups are still expected to be informative. Despite these limitations, the analysis provides meaningful insights into the challenge behavior with respect to game context and pitch location.

6 Conclusion & Future Work

In the decision to challenge a pitch, players take into account pitch location and the game context. Pitch location serves as the primary filter, as players generally reserve challenges for pitches near the edge of the strike zone. Game context determines whether it is worth using a limited challenge at that moment. The decision to implement a challenge is a balance between pitch accuracy and strategic considerations. ABS introduces a new strategy to MLB, making challenge decisions an important component of in-game decision making.

Future work should extend this analysis using data from the remainder of the 2026 MLB regular season and postseason. A larger dataset would make it possible to determine whether the observed trends persist throughout the season. Additionally, comparing regular season and postseason games could reveal whether teams adopt different challenge strategies in higher-leverage games. Finally, further analysis could compare challenge behavior across teams and identify which organizations have the most effective strategies.

References

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